

MA206 AY27-1 WPR I Review

Descriptive Statistics, Probability, and Random Variables

ANSWER KEY

On the WPR you are authorized the SRC, R-Lite, and your issued calculator. Give final answers to 3 decimal places.

Part A: Concept Checks

True/False, multiple choice, or fill in the blank. For every "False," be able to say why.

- True or False:** If $P(A) = 0.65$ and $P(B) = 0.55$, then A and B could be mutually exclusive.
False. If they were mutually exclusive, $P(A \cup B) = 0.65 + 0.55 = 1.2 > 1$, which violates the axioms.
- An analyst records the blood type (A, B, AB, O) of 40 Soldiers. This variable is:
(a) categorical (nominal) (b) categorical (ordinal)
(c) numerical, discrete (d) numerical, continuous
(a). Blood types are unordered categories.
- True or False:** A cumulative distribution function $F(x)$ can decrease as x increases.
False. $F(x) = P(X \leq x)$ can only stay the same or increase.
- Classify each study. (i) A researcher compares the GPAs of cadets who *choose* to drink energy drinks with those who do not. (ii) A PT instructor randomly assigns platoons to one of two stretching routines and compares injury rates.
(a) (i) observational study; (ii) designed experiment
(b) (i) designed experiment; (ii) observational study
(c) both are simple random samples
(d) both are censuses
(a). In (i) cadets chose the treatment, so confounding is possible and causation cannot be shown. In (ii) the instructor randomly assigned it, so (ii) can support a causal conclusion.
- A die is rolled. Let A = "the roll is even" and B = "the roll is greater than 3." The event $A \cap B'$ is:
(a) $\{2\}$ (b) $\{2, 4, 6\}$
(c) $\{4, 6\}$ (d) $\{1, 3\}$
(a). $B' = \{1, 2, 3\}$, and the only even number in it is 2.
- Fill in the blanks:** If $X \sim \text{Pois}(\mu = 6)$, then $E(X) = \underline{\hspace{1cm}}$, $V(X) = \underline{\hspace{1cm}}$, and $\text{SD}(X) = \underline{\hspace{1cm}}$.
6, 6, and $\sqrt{6} \approx 2.449$.
- True or False:** If the largest value in a sample is changed to a much larger value, the sample mean can change a great deal while the sample median does not change at all.

True. The mean uses every value, so it moves. The median depends only on the middle value(s), so it stays put.

8. **True or False:** For any events A and B , $(A \cup B)' = A' \cap B'$.

True (De Morgan). Not in A or B means not in A and not in B .

9. **True or False:** For a continuous random variable, $f(3)$ is the probability that $X = 3$.

False. A density value is not a probability; probabilities are areas, and $P(X = 3) = 0$.

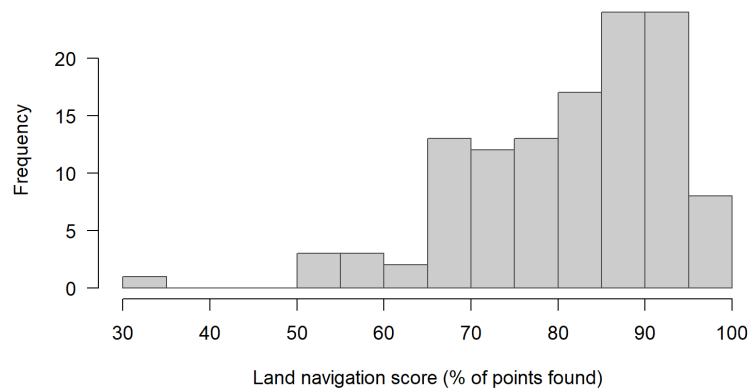
10. You roll a six-sided die and flip a coin. How many outcomes are in the sample space?

- (a) 6 (b) 8
(c) 12 (d) 36

(c). Product rule: $6 \times 2 = 12$.

11. The histogram below shows land navigation scores for 120 cadets. The shape is best described as:

- (a) right-skewed (b) symmetric
(c) left-skewed (d) uniform



(c). The long tail stretches toward the low scores (left).

12. For the histogram in the previous question, which relationship is most likely?

- (a) mean $>$ median (b) mean $\approx$ median
(c) mean $<$ median (d) cannot tell

(c). The tail on the left pulls the mean below the median. (Actual: mean = 81.7, median = 85.)

13. If $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, then $P(A \cap B)$ is:

- (a) 0.2 (b) 0.9
(c) 0.3 (d) 0.1

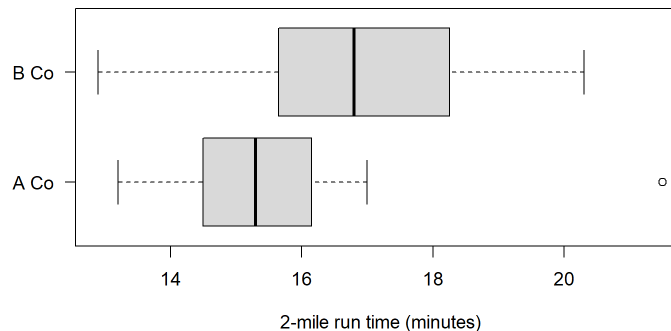
(a). $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.2$.

14. **Fill in the blank (memoryless property):** If T is exponential, $P(T > 5 \mid T > 2) = P(T > \text{---})$.

3. Having already waited 2 units does not change the chance of waiting 3 more.

15. **True or False:** The average fuel consumption of *every* vehicle in a brigade last month is a statistic.
False. It describes the entire population, so it is a parameter. A statistic is computed from a sample.
16. **True or False:** A squad of 11 Soldiers includes 4 medics. Select 5 Soldiers *without replacement*. Then X = the number of medics selected is binomial.
False. Without replacement from a small group, the trials are not independent: p changes after each pick.
17. **Fill in the blank:** If every value in a data set is multiplied by 3, the new sample variance is _____ times the original sample variance.
9. The standard deviation triples, so the variance is multiplied by $3^2 = 9$.
18. **Fill in the blank:** $P(A') = \underline{\hspace{2cm}}$.
 $1 - P(A)$.
19. A screening test detects a condition 95% of the time when it is present and gives a false alarm 5% of the time when it is absent. 1% of Soldiers have the condition. A Soldier tests positive. The probability they have the condition is closest to:
 (a) 0.95 (b) 0.50
 (c) 0.16 (d) 0.01
(c). $\frac{(0.95)(0.01)}{(0.95)(0.01) + (0.05)(0.99)} = \frac{0.0095}{0.059} \approx 0.161$. The condition is rare, so false alarms dominate.
20. **Fill in the blank:** If X is a continuous random variable, $P(X = 2.5) = \underline{\hspace{2cm}}$, so $P(X \leq 2.5)$ and $P(X < 2.5)$ are _____.
0; equal. A single point has zero area under the density.
21. The critical value $z_{0.05} = 1.645$ means:
 (a) $P(Z < 1.645) = 0.05$ (b) $P(Z > 1.645) = 0.05$
 (c) $P(|Z| > 1.645) = 0.05$ (d) $P(Z = 1.645) = 0.05$
(b). z_α has area α to its right.
22. A plant tests batteries from a production line that will keep producing batteries next year. The collection of all batteries of interest is best described as:
 (a) a finite population (b) a process (conceptual population)
 (c) a sample (d) a parameter
(b). The line keeps producing, so there is no fixed list of units: it is a process, or conceptual population.
23. **True or False:** The Poisson distribution models the *time* between arrivals.
False. The Poisson models the number of events in a fixed interval. The time between events is exponential.
24. **True or False:** There are more ways to choose 3 cadets from 10 than to choose 7 cadets from 10.
False. $\binom{10}{3} = \binom{10}{7} = 120$: choosing the 3 who go is the same as choosing the 7 who stay.

25. **True or False:** If 10 minutes is added to every run time in a data set, the standard deviation increases by 10.
False. Shifting every value moves the mean up by 10 but leaves the spread (and so s) unchanged.
26. Which of the following is a valid probability mass function on $x = 0, 1, 2$?
 (a) 0.3, 0.3, 0.3 (b) 0.5, 0.6, -0.1 (c) 0.25, 0.50, 0.25 (d) 0.4, 0.4, 0.4
(c). Each probability is between 0 and 1 and they sum to 1. (a) sums to 0.9; (b) has a negative value; (d) sums to 1.2.
27. $P(A) = 0.3$ and $P(B) = 0.5$. The events A and B are independent exactly when $P(A \cap B)$ equals:
 (a) 0.8 (b) 0.15
 (c) 0.2 (d) 0
(b). Independence means $P(A \cap B) = P(A)P(B) = 0.15$.
28. The comparative boxplots show 2-mile run times for two companies.



- (i) Which company has the larger median time? (ii) Which has the larger IQR?
 (iii) Does either company show an outlier?
(i) B Co (about 16.8 vs. 15.3). (ii) B Co (wider box). (iii) Yes: A Co has one outlier near 21.5 minutes.
29. The number of push-ups a Soldier completes in two minutes is:
 (a) categorical (nominal) (b) categorical (ordinal)
 (c) numerical, discrete (d) numerical, continuous
(c). It is a count.
30. **True or False:** For the standard normal random variable Z , $P(Z < -1.5) = P(Z > 1.5)$.
True. The standard normal curve is symmetric about 0.
31. **True or False:** Two mutually exclusive events, each with positive probability, are independent.
False. $P(A \cap B) = 0$, but $P(A)P(B) > 0$. Knowing A occurred tells you B did not, so they are dependent.
32. Which of the following is best modeled with a Poisson distribution?
 (a) the number of heads in 20 coin flips

- (b) the number of flat tires per 100 miles of convoy travel, at a steady average rate
 (c) the time until the next flat tire
 (d) the weight of a tire
(b). A count of events over a fixed interval at a steady rate. (a) is binomial, (c) is exponential, (d) is continuous.
33. **True or False:** A probability density function $f(x)$ can take values greater than 1.
True. Only the area must equal 1. For example, $f(x) = 2$ on $[0, 0.5]$ is a valid pdf.
34. **True or False:** In a simple random sample of size n , every possible sample of size n has the same chance of being selected.
True. That is the definition of a simple random sample.
35. **Fill in the blanks:** If $X \sim \text{Bin}(n, p)$, then $E(X) = \underline{\hspace{2cm}}$ and $V(X) = \underline{\hspace{2cm}}$.
 np and $np(1 - p)$.
36. **True or False:** Surveying the first 30 cadets who walk into the mess hall produces a simple random sample.
False. It is a convenience sample: not every group of 30 cadets had the same chance of being chosen.
37. **True or False:** If A and B are independent, then A' and B are also independent.
True. $P(A' \cap B) = P(B) - P(A)P(B) = P(B)[1 - P(A)] = P(A')P(B)$.
38. A pdf that is *decreasing* across its entire support describes a distribution that is:
 (a) left-skewed (b) symmetric
 (c) right-skewed (d) bimodal
(c). Most of the area is near the low end, with a tail stretching to the right, so mean > median.
39. A company commander summarizes her company's ACFT scores with a mean and a boxplot and draws no conclusions beyond her company. This is:
 (a) descriptive statistics (b) inferential statistics
 (c) a designed experiment (d) a census of the Army
(a). She is only describing the data in hand, not generalizing to a larger population.
40. If $T \sim \text{Exp}(\lambda = 0.08)$, the median of T is:
 (a) 12.5 (b) 8.664
 (c) 0.08 (d) 6.25
(b). $1 - e^{-0.08m} = 0.5 \Rightarrow m = \ln 2 / 0.08 \approx 8.664$, below the mean of 12.5.
41. **Fill in the blanks:** $E(aX + b) = \underline{\hspace{2cm}}$ and $V(aX + b) = \underline{\hspace{2cm}}$.
 $aE(X) + b$ and $a^2V(X)$. (Adding b does not change the variance.)
42. **True or False:** If $T \sim \text{Exp}(\lambda = 0.5)$, then the standard deviation of T is 0.5.
False. The mean and SD both equal $1/\lambda = 2$.
43. **True or False:** For any events A and B , $P(A | B) = P(B | A)$.
False. $P(A | B) = P(A \cap B)/P(B)$ and $P(B | A) = P(A \cap B)/P(A)$; they are equal

only when $P(A) = P(B)$.

44. Which of the following is *not* a requirement for a binomial experiment?

- (a) a fixed number of trials n (b) independent trials (c) the same p on every trial
- (d) the success probability must be $p = 0.5$

*(d). p can be any value between 0 and 1, as long as it stays the same on every trial.
(Each trial must also have two outcomes.)*

Part B: Quick Computations

Short calculations. Topics are mixed.

1. A vehicle bumper code is 3 letters followed by 2 digits, and repeats are allowed. How many codes are possible?

Product rule: $26^3 \times 10^2 = 1,757,600$.

The next three questions use these 9 observations: 3, 21, 22, 24, 25, 26, 27, 29, 38. Find Q_1 and Q_3 as the medians of the lower and upper halves, including the median in both halves when n is odd (Devore's fourths).

2. Find the sample mean and the sample median.

$\bar{x} = 215/9 \approx 23.889$; median = the 5th ordered value = 25.

3. Find Q_1 , Q_3 , and the IQR.

Lower half 3, 21, 22, 24, 25 $\Rightarrow Q_1 = 22$. Upper half 25, 26, 27, 29, 38 $\Rightarrow Q_3 = 27$. IQR = $27 - 22 = 5$.

4. Use the $1.5 \times$ IQR rule to identify any outliers.

Fences: $22 - 1.5(5) = 14.5$ and $27 + 1.5(5) = 34.5$. Both $3 < 14.5$ and $38 > 34.5$ are outliers.

5. $X \sim N(\mu = 50, \sigma = 8)$. Find $P(X < 62)$.

$z = (62 - 50)/8 = 1.50$; $P(Z < 1.50) = 0.9332$. (R-Lite: `pnorm(62, 50, 8)`.)

6. $X \sim N(\mu = 50, \sigma = 8)$. Find the 95th percentile of X .

$x = \mu + z_{0.95}\sigma = 50 + 1.645(8) \approx 63.160$. (R-Lite: `qnorm(0.95, 50, 8)`.)

7. $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$. Draw a Venn diagram and label the probability of all four regions. Then find $P(A' \cap B')$ and $P(A \cap B')$.

Regions: A only 0.4, both 0.1, B only 0.2, neither 0.3. $P(A \cup B) = 0.5 + 0.3 - 0.1 = 0.7$, so $P(A' \cap B') = 0.3$ and $P(A \cap B') = 0.4$.

8. The pdf of X is $f(x) = \frac{2(3-x)}{9}$ for $0 \leq x \leq 3$ (and 0 otherwise). Find $P(-1 \leq X \leq 1)$.

$F(x) = \frac{6x-x^2}{9}$ on $[0, 3]$ and $F(-1) = 0$ (the support starts at 0), so $P(-1 \leq X \leq 1) = F(1) = \frac{5}{9} \approx 0.556$.

9. Four components work independently, each with probability 0.95. Find the probability all four work.

$0.95^4 \approx 0.815$.

The next question uses this pmf:

x	0	1	2	3
$p(x)$	0.1	0.3	0.4	0.2

10. Find $E(X)$ and $V(X)$.

$E(X) = 0(.1) + 1(.3) + 2(.4) + 3(.2) = 1.7$; $E(X^2) = 0 + .3 + 1.6 + 1.8 = 3.7$; $V(X) = 3.7 - 1.7^2 = 0.81$.

11. Find the mean and the 10% trimmed mean of: 2, 14, 15, 16, 16, 17, 18, 19, 20, 22.

$\bar{x} = 159/10 = 15.9$. Trim 10% of $10 = 1$ from each end (drop 2 and 22): $135/8 = 16.875$.
The trimmed mean resists the 2.

12. The wait for a shuttle is exponential with a mean of 25 minutes. Find the probability of waiting more than 30 minutes.

$$\lambda = 1/25; P(T > 30) = e^{-30/25} = e^{-1.2} \approx 0.301.$$

13. A briefer must present 3 of 10 topics, and the order of presentation matters. How many briefing orders are possible?

$$P_{3,10} = 10 \cdot 9 \cdot 8 = 720.$$

14. $X \sim \text{Bin}(8, 0.2)$. Find $P(X = 2)$.

$$\binom{8}{2}(0.2)^2(0.8)^6 = 28(0.04)(0.262144) \approx 0.294. \quad (\text{R-Lite: } \text{dbinom}(2, 8, 0.2).)$$

15. $X \sim \text{Bin}(8, 0.2)$. Find $P(X \geq 1)$.

$$1 - P(X = 0) = 1 - 0.8^8 \approx 0.832.$$

16. Find the sample variance, sample standard deviation, and range of: 4, 7, 9, 10, 15.

$$\bar{x} = 9; \sum (x_i - \bar{x})^2 = 25 + 4 + 0 + 1 + 36 = 66; s^2 = 66/4 = 16.5; s = \sqrt{16.5} \approx 4.062; \text{range} = 15 - 4 = 11.$$

17. Aircraft land at an airfield at an average rate of 1.5 per hour. Find the probability of no landings in a 3-hour window.

$$Y \sim \text{Pois}(\mu = 1.5 \cdot 3 = 4.5); P(Y = 0) = e^{-4.5} \approx 0.011.$$

18. $P(A \cap B) = 0.12$ and $P(B) = 0.40$. Find $P(A | B)$.

$$P(A | B) = 0.12/0.40 = 0.30.$$

19. A section has 5 women and 7 men. Four cadets are chosen at random for a briefing team. What is the probability exactly 2 are women?

$$\frac{\binom{5}{2}\binom{7}{2}}{\binom{12}{4}} = \frac{10 \cdot 21}{495} \approx 0.424.$$

20. In a battalion, 30% of Soldiers are Airborne qualified, 40% Air Assault qualified, and 10% Ranger qualified. 15% hold both Airborne and Air Assault, 8% Airborne and Ranger, 7% Air Assault and Ranger, and 5% hold all three. Find the probability a randomly chosen Soldier holds at least one of the three.

$$0.30 + 0.40 + 0.10 - 0.15 - 0.08 - 0.07 + 0.05 = 0.55.$$

Part C: Full Problems

Define your random variable before computing.

Problem 1. Let X be the number of cadets in a squad who are late to morning formation, with pmf

$$p(x) = \begin{cases} 0.10 & x = 0 \\ 0.25 & x = 1 \\ 0.30 & x = 2 \\ 0.20 & x = 3 \\ 0.15 & x = 4 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cdf $F(x) = P(X \leq x)$ and write it in full piecewise form.

$$\text{Add the pmf as you go: } F(x) = P(X \leq x) = \begin{cases} 0 & x < 0 \\ 0.10 & 0 \leq x < 1 \\ 0.35 & 1 \leq x < 2 \\ 0.65 & 2 \leq x < 3 \\ 0.85 & 3 \leq x < 4 \\ 1 & x \geq 4. \end{cases}$$

(b) Find $P(X < 2)$.

$$P(X < 2) = P(X \leq 1) = F(1) = 0.35.$$

(c) Find $P(X \leq 2)$.

$$F(2) = 0.65. \text{ It differs from (b) by } p(2) = 0.30.$$

(d) Find $P(X > 3)$.

$$1 - F(3) = 1 - 0.85 = 0.15.$$

(e) Find $P(X \geq 3)$.

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - F(2) = 0.35.$$

(f) Find μ and σ . Then find the probability X is within one standard deviation of its mean.

$$\text{From the pmf: } \mu = 0(.10) + 1(.25) + 2(.30) + 3(.20) + 4(.15) = 2.05; E(X^2) = 5.65; \\ V(X) = 5.65 - 2.05^2 = 1.4475; \sigma \approx 1.203. \mu \pm \sigma \approx (0.847, 3.253), \text{ so } P(0.847 < X < 3.253) = P(1 \leq X \leq 3) = F(3) - F(0) = 0.85 - 0.10 = 0.75.$$

Problem 2. The time X (in hours) a cadet spends on homework in an evening has pdf

$$f(x) = \begin{cases} \frac{x}{8} & 0 \leq x \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cdf $F(x) = P(X \leq x)$ and write it in full piecewise form.

$$\text{For } 0 \leq x \leq 4: F(x) = \int_0^x \frac{t}{8} dt = \frac{x^2}{16}. \text{ So } F(x) = P(X \leq x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{16} & 0 \leq x \leq 4 \\ 1 & x > 4. \end{cases}$$

- (b) Find $P(X < 2)$.
 $F(2) = \frac{4}{16} = 0.25$.
- (c) Find $P(X \leq 2)$.
 $F(2) = 0.25$, the same as (b), because $P(X = 2) = 0$ for a continuous random variable.
- (d) Find $P(X > 3)$.
 $1 - F(3) = 1 - \frac{9}{16} = 0.4375$.
- (e) Find $P(X \geq 3)$.
 $1 - F(3) = 0.4375$, the same as (d).
- (f) Find $P(X = 3)$.
 0. For a continuous random variable, a single value has no area under the pdf. That is why (d) and (e) are equal.
- (g) Find μ and σ . Then find the probability X is more than one standard deviation above its mean.
 $\mu = \int_0^4 x \cdot \frac{x}{8} dx = \frac{8}{3} \approx 2.667$; $E(X^2) = \int_0^4 x^2 \cdot \frac{x}{8} dx = 8$; $V(X) = 8 - \frac{64}{9} = \frac{8}{9} \approx 0.889$;
 $\sigma \approx 0.943$. $\mu + \sigma \approx 3.609$, so $P(X > 3.609) = 1 - F(3.609) = 1 - \frac{3.609^2}{16} \approx 0.186$.

Problem 3. Two-mile run times X for a company are normal with mean 15.5 minutes and standard deviation 1.2 minutes.

- (a) Find $P(X < 14)$.
 $z = (14 - 15.5)/1.2 = -1.25$; $P(Z < -1.25) = 0.1056$. (R-Lite: `pnorm(14, 15.5, 1.2)`.)
- (b) Find $P(X \leq 14)$.
 0.1056, the same as (a) (continuous).
- (c) Find $P(X > 18)$.
 $z = (18 - 15.5)/1.2 \approx 2.08$; $P(Z > 2.08) = 1 - 0.9812 = 0.0188$. (R-Lite: `1 - pnorm(18, 15.5, 1.2)` ≈ 0.019 .)
- (d) Find $P(X \geq 18)$.
 0.0188, the same as (c).
- (e) Find $P(X = 18)$.
 0. For a continuous random variable, a single value has no area under the pdf. That is why (c) and (d) are equal.
- (f) Find μ and σ . Then find the probability X is more than two standard deviations above its mean.
 $\mu = 15.5$, $\sigma = 1.2$. $\mu + 2\sigma = 17.9$, so $P(X > 17.9) = P(Z > 2) = 1 - 0.9772 = 0.0228$.
 This is the same for every normal distribution.

Problem 4. Each of 15 Soldiers independently qualifies Expert at the range with probability 0.3. Let X be the number who qualify Expert, so $X \sim \text{Bin}(15, 0.3)$.

- (a) Find $P(X < 4)$.
 $P(X < 4) = P(X \leq 3) = F(3) \approx 0.297$. (R-Lite: `pbinom(3, 15, 0.3)`.)
- (b) Find $P(X \leq 4)$.
 $F(4) \approx 0.515$. (R-Lite: `pbinom(4, 15, 0.3)`.)
- (c) Find $P(X > 6)$.
 $1 - P(X \leq 6) = 1 - F(6) \approx 0.131$. (R-Lite: `1 - pbinom(6, 15, 0.3)`.)
- (d) Find $P(X \geq 6)$.
 $1 - P(X \leq 5) = 1 - F(5) \approx 0.278$. (R-Lite: `1 - pbinom(5, 15, 0.3)`.)
- (e) Find μ and σ . Then find the probability X is more than two standard deviations below its mean.
 $\mu = 15(0.3) = 4.5$; $\sigma = \sqrt{15(0.3)(0.7)} = \sqrt{3.15} \approx 1.775$. $\mu - 2\sigma \approx 0.950$, so $P(X < 0.950) = P(X = 0) = 0.7^{15} \approx 0.005$. (R-Lite: `dbinom(0, 15, 0.3)`.)

Problem 5. Calls to the staff duty desk arrive randomly and independently at an average rate of 6 per hour. Let X be the number of calls in one hour.

- (a) What distribution does X follow?
 $X \sim \text{Pois}(\mu = 6)$: a count of events in a fixed interval at a steady rate.
- (b) Find $P(X < 5)$.
 $P(X < 5) = P(X \leq 4) = F(4) \approx 0.285$. (R-Lite: `ppois(4, 6)`.)
- (c) Find $P(X \leq 5)$.
 $F(5) \approx 0.446$. (R-Lite: `ppois(5, 6)`.)
- (d) Find $P(X > 8)$.
 $1 - P(X \leq 8) = 1 - F(8) \approx 0.153$. (R-Lite: `1 - ppois(8, 6)`.)
- (e) Find $P(X \geq 8)$.
 $1 - P(X \leq 7) = 1 - F(7) \approx 0.256$. (R-Lite: `1 - ppois(7, 6)`.)
- (f) Find μ and σ . Then find the probability X is within two standard deviations of its mean.
 $\mu = 6$; $\sigma = \sqrt{6} \approx 2.449$. $\mu \pm 2\sigma \approx (1.101, 10.899)$, so $P(2 \leq X \leq 10) = F(10) - F(1) \approx 0.940$. (R-Lite: `ppois(10, 6) - ppois(1, 6)`.)

Problem 6. Return to the staff duty desk in Problem 5, with calls arriving at an average rate of 6 per hour. Now let T be the time (in minutes) between one call and the next.

- (a) What distribution does T follow? Write its pdf. Then find the cdf $F(t) = P(T \leq t)$ and write it in full piecewise form.
The time between Poisson events is exponential. 6 per hour is $\lambda = 0.1$ per minute (mean 10 minutes), so $f(t) = 0.1e^{-0.1t}$ for $t \geq 0$ (and 0 otherwise). For $t \geq 0$:

$$F(t) = \int_0^t 0.1e^{-0.1s} ds = 1 - e^{-0.1t}$$
. So $F(t) = P(T \leq t) = \begin{cases} 0 & t < 0 \\ 1 - e^{-0.1t} & t \geq 0. \end{cases}$

- (b) Find $P(T < 6)$.
 $F(6) = 1 - e^{-0.6} \approx 0.451$. (R-Lite: `pexp(6, 0.1)`.)
- (c) Find $P(T \leq 6)$.
 $F(6) \approx 0.451$, the same as (b) (continuous).
- (d) Find $P(T > 15)$.
 $1 - F(15) = e^{-1.5} \approx 0.223$. (R-Lite: `1 - pexp(15, 0.1)`.)
- (e) Find $P(T \geq 15)$.
 $1 - F(15) \approx 0.223$, the same as (d).
- (f) Find $P(T = 15)$.
 0. For a continuous random variable, a single value has no area under the pdf. That is why (d) and (e) are equal.
- (g) Find μ and σ . Then find the probability T is within one standard deviation of its mean.
 $\mu = \sigma = 1/\lambda = 10$ minutes. $\mu \pm \sigma = (0, 20)$, so $P(0 < T < 20) = F(20) - F(0) = 1 - e^{-2} \approx 0.865$. This is the same for every exponential distribution. (R-Lite: `pexp(20, 0.1)`.)

Problem 7. In a motor pool, 60% of vehicles are wheeled and 40% are tracked. 10% of wheeled vehicles and 25% of tracked vehicles fail their monthly inspection. A vehicle is selected at random.

- (a) Find the probability the vehicle fails inspection.
 Let $W = \text{wheeled}$, $T = \text{tracked}$, $F = \text{fails inspection}$. Given: $P(W) = 0.6$, $P(T) = 0.4$, $P(F | W) = 0.10$, $P(F | T) = 0.25$.
 Law of Total Probability: $P(F) = P(F | W) P(W) + P(F | T) P(T) = (0.10)(0.6) + (0.25)(0.4) = 0.06 + 0.10 = 0.16$.
- (b) The vehicle failed inspection. What is the probability it is tracked?
 Bayes' Theorem: $P(T | F) = \frac{P(F | T) P(T)}{P(F | W) P(W) + P(F | T) P(T)} = \frac{(0.25)(0.4)}{(0.10)(0.6) + (0.25)(0.4)} = \frac{0.10}{0.16} = 0.625$.